

Routing Algorithms

1 Introduction to Routing Algorithms

Routing Algorithm

Goal: To determine "good" paths (routes) from sending hosts to receiving hosts through a network of routers.

What defines a "Good" Path?

- **Least Cost:** Typically the path with the minimum total cost (distance, speed, or \$.).
- **Policy Issues:** Real-world constraints (e.g., "Router X should not forward packets from Organization Y").

2 Graph Abstraction

To formulate routing problems, we use a graph model $G = (N, E)$.

The Components

- **Nodes (N):** Represent routers (or networks in BGP).
- **Edges (E):** Represent physical links between routers.
- **Costs ($c(x, y)$):** Each edge has a value representing length, speed, or monetary cost.

2.1 Path Cost calculation

Least-Cost Path

A **path** is a sequence of nodes $(x_1, x_2, \dots, x_p)$.

The **cost of a path** is the sum of all edge costs:

$$Cost(x_1, x_2, \dots, x_p) = c(x_1, x_2) + c(x_2, x_3) + \dots + c(x_{p-1}, x_p)$$

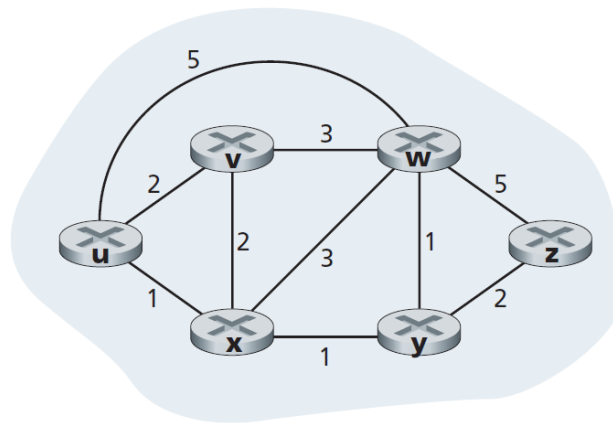


Figure 1: Abstract graph model of a computer network

Shortest Path vs. Least-Cost

If all edges have the **same cost**, the least-cost path is simply the **shortest path** (minimum number of hops).

3 Routing Algorithm Classification

Routing algorithms are primarily classified according to their **information scope** and their **speed of adaptation**.

Criterion 1: Information Scope (Global vs. Decentralized)

This classification asks: *"Does the algorithm have a complete view of the network or only local information?"*

- **Global ("Link State" Algorithms):**

- **Knowledge:** All routers have **complete topology** and **link cost information** for the entire network.
- **Input:** The connectivity between all nodes and all link costs must be obtained before calculation.
- **Example:** OSPF.

- **Decentralized ("Distance Vector" Algorithms):**

- **Knowledge:** Distributed and iterative. No node has complete information about all link costs.
- **Process:** Routers initially only know link costs to their **directly attached neighbors**.
- **Calculation:** Gradual calculation via an **iterative process** involving information exchange with neighbors.

- **Example:** RIP.

Criterion 2: Change Speed (Static vs. Dynamic)

This classification asks: *"How fast do the routes change over time?"*

- **Static Routing:**

- **Behavior:** Routes change very slowly over time.
- **Intervention:** Typically requires **human intervention** (manual editing of link costs or tables).

- **Dynamic Routing:**

- **Behavior:** Routes change more quickly and automatically.
- **Triggers:** Changes occur periodically or in direct response to **topology changes** or **link cost changes**.
- **Trade-off:** More responsive but susceptible to routing loops and oscillations.

4 Routing Protocols Hierarchy

Modern networks categorize routing protocols based on their scope (Autonomous Systems) and underlying algorithm.

Intra-domain vs. Inter-domain

- **Intra-domain (IGP - Interior Gateway Protocols):** Used within a single Autonomous System (AS). Focused on efficiency and speed.
- **Inter-domain (EGP - Exterior Gateway Protocols):** Used between different Autonomous Systems. Focused on policy, security, and scalability.

4.1 Intra-domain Routing Protocols

Used to route packets within a local network or organization.

Examples of IGP

- **RIP (Routing Information Protocol):**
 - **Type:** Distance Vector (DV).
 - **Metric:** Hop count (max 15).
 - **Update:** Periodic exchange with neighbors.
- **OSPF (Open Shortest Path First):**
 - **Type:** Link State (LS).

- **Metric:** Cost (usually bandwidth).
- **Update:** Floods link-state advertisements (LSAs).

4.2 Inter-domain Routing Protocol

Used to connect various Autonomous Systems (the Internet's backbone).

The Standard: BGP

BGP (Border Gateway Protocol):

- **Type: Path Vector** (extension of DV).
- **Purpose:** Enables routing between ISPs.
- **Focus:** Policy-based routing.

5 Key Takeaways

Summary

- **LS vs DV:** Global vs. Distributed calculation.
- **Static vs Dynamic:** Manual vs. Automated updates.
- **IGP vs EGP:** Intra-domain (RIP, OSPF) vs. Inter-domain (BGP).